27. Which of the following has a positive effect on our local vegetation?
A. Random introduction of exotic plants.
B. Replanting the land with endemic species.
C. Production of timber for commercial purpose.
D. Rapid harvesting without planting.

Explanation:
Replanting the land with endemic species has a positive effect on local vegetation. **Answer: B**

28. Which of the following agrees with the basic principles of vegetation conservation?
A. Using the vegetation through proper conservation.
B. Introduction of exotic plants.
C. Removing all the wild animals naturally living in a vegetation.
D. Planting the land only with endemic plants.

Explanation: Refer example number 27. **Answer: A**

29. Vegetation is used for all of the following except one:
A. Source of food
B. Stabilization of atmospheric oxygen and carbon dioxide
C. Increasing the rate of pollutants release
D. Habitat for variety of organisms

Explanation: Vegetation do not play role in increasing pollutants in the atmosphere. **Answer: C**

30. One of the following is NOT recommended as a means of rehablitation of degraded natural vegetation:
A. Reforestation
B. Control of fire
C. Public education
D. Free grazing by domestic animal

Explanation:
Degraded natural vegetation can rehabilitated through:
- Reforestation
- Control of fire
- Public education
- Avoid grazing by domestic animals **Answer: D**

31. Which of the following plants is NOT endemic to Ethiopia?
A. Zigba
B. Maize
C. Teff
D. Enset

Explanation: Endemic plants are plants that are found only in a given country. Some of the endemic plants in Ethiopia include zigba, teff, enset, etc. However, maize is not endemic to Ethiopia.

Answer: B

32. How do predators contribute in conserving our vegetation?
A. By feeding on a specific species of plants.
B. By keeping down the number of herbivores.
C. By feeding on unimportant weeds.
D. By eradicating herbivores from an area.

Explanation: Predators play role by eating or hunting herbivores animals that may overgraze the vegetation. Thus, existence of predators is important for keeping balance in an ecosystem.

Answer: B

33. One negative global consequence of large scale deforestation is the one that may result in:
A. A rise in the atmospheric composition of N_2.
B. A rise in the atmospheric composition of O_2.
C. A rise in the atmospheric composition of CO_2.
D. A reduction in the atmospheric composition of CO_2.

Explanation:
Deforestation results in the rise of atmospheric CO_2. This is due to the reason that if plants are deforested the amount of atmospheric CO_2 absorbed by plants for photosynthesis declines. This rises atmospheric CO_2.

Answer: C

34. What do we mean when we say an organism is endemic to Ethiopia?
A. First discovered in Ethiopia.
B. Found in large numbers in Ethiopia.
C. Endangered in Ethiopia.
D. Found only in Ethiopia.

Explanation: Endemic species in a species that is only found in one country.

Answer: D

35. One of the following activities is used in conserving vegetation. Which one is it?
A. Destruction of forest habitat.
B. Planting indegineous trees.
C. Introducing exotic plants in large scale.
D. Hunting of wild animals living in vegetation.

Explanation: For conservation of vegetation, the following should be done:

- Protection and preservation of forest habitat
- Planting indegineosu trees
- Protection of wild life living in vegetation

Answer: B

36. Which of the following plant is endemic to Ethiopia?

A. Eucalyptus tree C. Enset ventricusum
B. Maize D. Banana

Explanation: Except Enset, none of the plants listed are endemic to Ethiopia. **Answer: C**

37. List out the appropriate majors that should be taken to preserve Ethiopian vegetation.

Explanation: For preservation of vegetation at any place including our country the following should be done:

- ✓ Planting trees
- ✓ Providing protection for the planted trees
- ✓ Educating the society

38. What are endemic plants? List out some examples of endemic plants in Ethiopia.

Explanation: Endemic plants are plants that grow in one country only. Some examples endemic plants in Ethiopia are:

- Teff	- Ficus vista	- kerero
- Noug	- Zigba	- Sembo
- Enset	- juniper	- Anchote

39. Distinguish between indegineous and exotic plants. List out some examples for each under the context of Ethiopia.

Explanation:

i) **Exotic plants** are plants that are not originally from one locality, but are brought from other places having their own climate and topography.

Examples: In Ethiopia plants such as Bahirzaf (Eucalyptus) are exotic.

ii) **Indigenosu plants** – are plants that originally grown in a given locality or country. They are highly adapted to the climatic and topography of an area.

Examples: Teff, Wanza, Kerero etc

40. What is the significance of planting trees? To
A. reduce the number of wildlife.
B. reduce the number of soil microorganism.
C. reduce the rate of soil erosion.
D. increase the amount of soil erosion.

Explanation: Planting trees has several significances. These are:
- ✓ increase the number of wild life
- ✓ enhance the number of soil microorganism.
- ✓ reduce the rate of soil erosion.
- ✓ recue the amount of CO_2 in the air. **Answer: C**

41. Which of the following statement is true about vegetation? It
A. Reduces the amount of O_2 in the air.
B. Causes global warming.
C. Causes air pollution.
D. Is a source of clothes and foods.

Explanation: Vegetations
- Rises the amount of CO_2 in the air
- Reduces global warming
- Reduces air pollution
- Used in making buildings, clothes, foods etc. **Answer: D**

42. Which of the following plant is endemic to our country?
A. Coffee C. Bean
B. Maize D. Sembo

Explanation: Coffee, maize and bean are not endemic to Ethiopia. **Answer: D**

Post Test – 2

Part I: Choose the best answer from the given alternatives

1. How do predators contribute in conserving our vegetation?
A. By feeding on a specific species of plants.
B. By eradicating herbivores form an area.
C. By keeping down the number of herbivores.
D. By feeding on unimportant weeds

2. Which of the following activities is used to conserve our vegetation?
A. Protecting natural resources.
B. Planting exotic plants.
C. Reducing indigenous plants.
D. Encouraging deforestation

3. Which of the following plants is endemic to our country?

A. Coffee B. Maize C. Bean D. Sembo

4. What is the importance of vegetation? It can be used:
   A. as a source of food
   B. as a source of acid rain.
   C. for balancing of sulfourdioxide.
   D. for balancing of nitrogen oxide
5. Which of the following is a method of vegetation?
   A. replanting of lands with indeginous trees.
   B. Burning of plants which have no immediate use
   C. Planting trees that take long time to produce seeds
   D. Covering more areas of land with exotic plants.
6. Which of the following statement is true about vegetation? It:
   A. reduces the amount of O_2 in the air.
   B. causes global warming.
   C. causes air pollution.
   D. is a source of clothes and foods.
7. Which of the following has positive effect on vegetation?
   A. Deforestation C. Reforestation
   B. Overgrazing D. Monoculture

Part II. Give short answer for the following items.

8. List out at least four endemic animals found Ethiopia
9. Describe the conservation mechanisms to preserve Ethiopia vegetation.
10. What are the uses of vegetation for humans? List out some of them

11.4 Wildlife

Wildlife refers to living things and especially mammals, birds, and fishes that are neither human nor domesticated. They can be found in forest, desert, or anywhere else.

Ethiopia is one of the richest countries in its wildlife resource. This involves variety of mammals, birds, insects and reptiles, etc.

Wild life reserve has several economic, educational and ecological importances.

These include:

- ***Economic***: revenue from tourism.

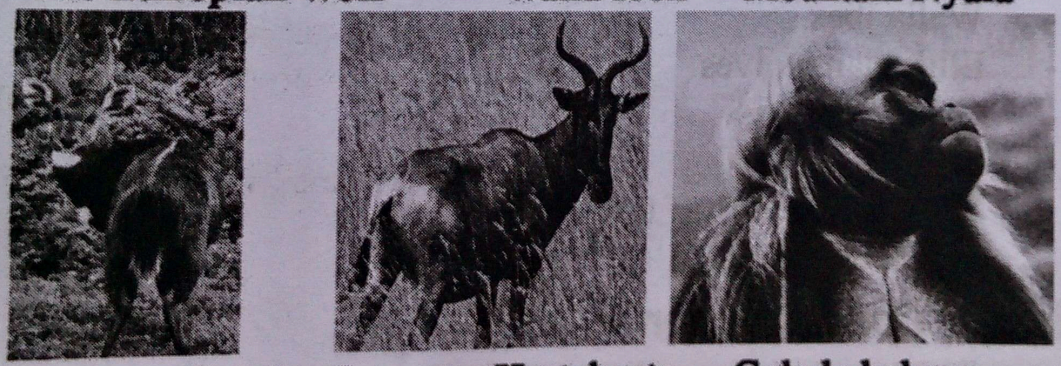
- ***Educational:*** for further scientific studies.
- Recreational value
- ***Ecological***: maintenance of natural balance in the ecosystem. For instance if we look at the predators like lion and tiger in an ecosystem, they keep down the number of herbivores which will largely help to maintain the balance in an ecosystem.

Endemic species of wild life in Ethiopia

In Ethiopia there are several types of wild animals. These include:

- Gelada baboon (ጭላዳ ዝንጀሮ)
- Walia ibex (ዋልያ)
- Menelek's bushbuck (የሚኒልክ ድኩላ)
- Mountain nyala (የደጋ አጋዘን)
- Swyne's hartebeest (የስዋይኔ ቆርኪ)
- Ethiopian wolf (ቀበሮ)

The Ethiopian Wolf Walia Ibex Mountain Nyala



Menelek Bushbuck Swaynes Hertebeste Gelade baboon

Figure 11.1: Endemic Animals in Ethiopia

Endemic Bird Species in Ethiopia

In Ethiopia there are about species of birds. Among these here are some of them:

- Thick billed raven (ቁራ)
- Wattled Ibis (ጋጋኖ)

- Black winged lovebird
- White collared pigeon
- Prince Ruspolis Turaco

Human Effect on Wildlife

The wild life reserve can be largely affected by several factors. Some of them are natural and catastrophic; while other are manmade (human impacts).

The human impacts are under the control of human beings if a proper conservation and protection mechanisms are implemented.

Some of the human effects on wildlife are:

- Destruction of forest habitat (deforestation) which is a home for wild animals.
- Uncontrolled hunting which is done mostly for using animals a source of food, for trading, for pride, for clothing.
- Human settling in a wild habitat.

Therefore, human activities keep down and reduce the number of wild animals which finally leads to extinction. Accordingly, some of the wild species of Ethiopia are in a danger of getting extinct.

Some of the Ethiopian wild species that are on the brick of extinction are:

- Walia ibex
- Ethiopian wolves
- Grevy's zebras
- Mountain nyala
- White winged fluff tail
- Ankober serin bird

Wildlife Conservation

Wildlife conservation involves the protection of habitat and managing populations. This is done through:

- Establishment of wildlife conservation areas such as national parks, sanctuaries, zoos, wildlife reserves.
- Preventing the spread of diseases. For instance vaccination of Ethiopian wolves against a viral disease called **rabies**.
- Controlling and managing human settlement in wild habitats.
- Establishing laws to control illegal acts or activities on wildlife. i.e. hunting, habitat destruction, etc..

- Awareness creation in the society.

Wildlife Conservation Areas

As mentioned above, conservation of wildlife is a core in keeping our global ecosystem stable. Accordingly, wildlife can be conserved by establishing wildlife conservation areas. Some of these include:

- ✓ ***National parks:*** are conservation areas designed to give maximum protection to wild animals.
- ✓ ***Wildlife sanctuaries***: are conservation areas established to give emphasis for a particular species.
  Example: Babile wildlife sanctuary.
- ✓ ***Wildlife reserves***: is an area for conservation, management and propagation of wildlife.
  - It involves controlling human settlement, hunting and capturing animals.
  - It is a highly protected area.
- ✓ ***Zoo***: is a facility in which all animals are housed within enclosures, displayed to the public, and in which they may also breed.

Table: National parks in Ethiopia

National park	Area	Location	Wild animals
Ablijatta – shalla lakes	887km^2	Oromia	Flamingos, white pelicans Golden Jackals, oribi warthog, Grant's Gazelle
Awash	786 km^2	Oromia	Wild pigs, zebra, dik – dik, Anubis and Hamadyras Baboons, cheetahs, leopards, ostriches, carmine beeeaters, Abyssinian roller
Bale mountains	2,150 km^2	Oromia	Gelada Baboon, mountain Nyala, Ethiopian Wolves, Minellik's Bush buck, etc.
Gambela	5000 km^2	Gambella	Nile perch, crocodiles, hippos, Roan antelope, Hyena, Lions, elephants, buffalo, zebra, vervet monkeys and black and white monkeys.

Riftvally lakes			Flamingos, white pelicans, fish eagles, speoonbills, abdim's stprls, ibises. Mammals like Warthogs Grant's Gazelle.
Mago	2000 km^2	SNNPR	Giraffe, elephants, lions, buffalo, cheetah, zebra, leopard, oxyx, vultures, Black rhinos
Omo	4000 km^2	SNNPR	Kudu, hartebeest, oryx, Anubis Baboons, lions, cheetah, buffalo, graffes, elephants
Semien Mountain	220 km^2	Amhara	Walia ibex, Ethiopian wolves, Gelada Baboon
Yangudi Rassa	4730 km^2	Afar	Wildass, kudu, Grevy's Zebra, cheetah
Maze	210 km^2	SNNPR	Swayne hartbeast

Illustrative Example

43. The cause of extniction and reduction in the number of wildlife can be:
    A. Uncontrolled hunting
    B. Alteration of natural habitat
    C. Overgrazing by domestic stock.
    D. Use of chemical pesticides.
    E. All are possible

    Explanation: Extrication and reduction in the number of wildlife can be caused by:
    - Uncontrolled hunting
    - Change in natural habitat
    - Overgrazing
    - Chemical pesticides, etc.

    Answer: E

44. Which of the following would be wise to adopt as a strategy of conserving Ethiopian wildlife?
    A. Selective breeding
    B. Selective killing of young animals.
    C. Introducing natural predators to control rising numbers of herbivores.
    D. Selective killing of old males.

    Answer: B

45. In which of the following alternatives are the animals endemic to Ethiopia?
A. Walia ibex, spotted hyena, zebra
B. Giraffe, zebra, camel
C. Thick billed raven, wattled ibis, elephant
D. Mountain Nyala, Walia ibex, Semien Fox.

Explanation: Endemic animals of Ethiopia include:
- Walia ibex
- Thick billed raven
- Nattled ibis
- Mountain Nyala
- Semien Fox

Answer: D

46. Walia ibex is to Semien National Park as _____is to Bale mountains National Park.
A. Thick billed Reven
B. Gelada Baboon
C. Mountain Nyala
D. Oryx

Explanation:

Endemic species	Located in
Thick billed (kura)	Central highlands of Ethiopia
Gelada baboon	Semein mountain and Menz
Mountain Nyala	Bale mountains

Answer: C

47. Which of the following is NOT an Ethiopian endemic animal?
A. Semien fox
B. Mountain Nyala
C. Hamdryas Baboon
D. Wattled Ibis

Explanation: Hamdryas Baboon is Not Ethiopian Endemic animal.

Answer: C

48. Which of the following national park is found in Oromia region?
A. Bale mountain
B. Omo
C. Mago
D. Chebera – churchura

Explanation:

National Park	Location
Bale mountain	Oromia region
Omo	SNNPR
Mago	SNNPR
Chebera – churchura	SNNPR

Answer: A

49. The largest national park in Ethiopia is:

A. Abijatta shala lakes national park.
B. Awash national park.
C. Bale mountain national park.
D. Gambella national park.

Explanation: Ethiopian national parks listed and their total area can be indicated as:

National park	Total area
Gambella national park	5000 km^2
Abjatta-shala lakes national park	887 km^2
Awash national park	756 km^2
Bale mountains national park	2150 km^2

Answer: D

50. Suppose a tourist is interested to visit variety of species of birds in Ethiopia. To which national park would you recommend him to visit?
A. Semien mountain
B. Bale mountain
C. Rift Valley lakes
D. Gambella national park

Explanation: Rift Valley lakes national park consists of various types of bird species like flamingos, white pelicans, fish, enagles, spoon bills, etc.

Answer: C

51. Why should we protect endemic animals such as Walia ibex from hunters?
A. It is an endemic antelope
B. It has an economic value.
C. It has the right to exist in its habitat.
D. All of the above

Answer: D

52. An animal or plant species restricted in the political boundary of a given country is known as:
A. Extinct species
B. Endangered species
C. Indigenous species
D. Endemic species

Explanation:

- **Extinct species** – a species that is completely eliminated from the earth by the natural hazards or human activities.
- **Endangered species** – a species which number is declining at an alarming rate and are on the verge of extinction.
- **Indigenous species** – a species (plant or animal) originating or native to country.

- **Endemic species** – plant or animal species found only in a particular geographical location or country.

53. Which of the following animals is found in Bale mountains national park? **Answer: D**

A. Mountain Nyala
B. Zebra
C. Flamingo
D. Walya

Explanation: Bale mountains national Park is a home for the following wild animals:

- Gelade Baboon
- Mountain Nyala
- Ethiopian Wolves
- Minelik's Bush buck

54. Which of the following animal is registered to be on the brick of extinction in Ethiopia? **Answer: A**

A. Mountain Nyala
B. Buffalo
C. Gelada baboon
D. Cheetah

Explanation: Endemic animals in Ethiopia that are on the brick of extinction are:

- Walia ibex
- Ethiopian wolves
- Greby's zebras
- Mountain Nyala
- White winged flufftail
- Ankober serin bird

Answer: A

55. Wildlife conservation include all of the following except:

A. Establishing national par
B. Hunting
C. Protection of habitat
D. Appropriate laws on wildlife management

Explanation:

Wildlife conservation involves protection and preservation of wild life through:

- Establishing national parks, zoos, sanctuaries, etc.
- Protection of natural habitat
- Setting appropriate laws and legal action **Answer: B**

56. The difference between sanctuaries and national parks lies in that, sanctuaries:

A. give more attention to vegetation than wildlife conservation.
B. are limited to highland area of our country.
C. are devoted to the conservation of particular species of wild life.
D. do not have protected boundaries.

Explanation:

National parks	Sanctuaries
- Designed to give maximum protection to the largest wild animals	Devoted to the conservation of a particular species of wild life

Answer: C

57. The conservation area which is designed to give maximum protection to the largest wild animals is:
A. Sanctuary
B. National Park
C. Wild life museum
D. Wild life reserve

Explanation: Refer above example. **Answer: B**

58. All are conserved in Semien mountain except:
A. Buffalo
B. Gelada baboon
C. Walia
D. Ethiopian Wolf

Explanation: Semien mountains national park is a home for:
- Walia ibex
- Ethiopian Wolves
- Gelabon Baoon

Answer: A

59. Rearrange the following national parks according to increasing order in their size (area).
i) Awash National park
ii) Abijatta Shalla lakes national park
iii) Bale Mountaisn National park
iv) Gambella National Park
v) Semien mountain national park

Explanation:

National Park	Area in km^2
Semien mountain	$220km^2$
Awash	$786\ km^2$
Abijatta Shala	$887\ km^2$
Bale mountains	$2150\ km^2$
Gambella	$5000\ km^2$

60. List out some of the endemic animals and bird species in Ethiopia with their locations.

Explanation:

Endemic animal	Location
Gelada Baboon	Bale Mountains, Semien Mountains
Walia ibex	Semien mountains
Meneliks Bushbuck	Bale mountains
Swayne's Hartbeest	Maze national park
Ethiopian Wolf	Bale mountains, Semien mountains
Thick billed raven	Western and southern highlands
Walted ibis	Ethiopian platoes
Blackwinged love bird	Rift valley and highlands
White collar pigeon	Western and southern highlands
Prince ruspolis turaco	Southern Ethiopia

Part Test – 3

Part I: Choose the best answer from the given alternatives.

1. Which of the following is the major reason to allow hunting in national parks?
   A. To raise huge amount of money needed for the park.
   B. To remove damage on the ecosystem.
   C. To get rid of wild animals those are dangerous to tourism.
   D. To fulfill the meat demand of the country.
2. Which of the following Ethiopian national parks has been declared as a World heritage site?
   A. Semien mountains National Park.
   B. Awash National Park.
   C. Bale Mountain national Park.
   D. Mago national park.
3. Which of the following animals is found in Bale mountains national Park?
   A. Mountain Nyala
   B. Zebra
   C. Flamingo
   D. Walya
4. Which one of the following is endemic bird species in Ethiopia?
   A. Gelada Baboon
   B. Walia ibex
   C. Wattled ibis
   D. Ethiopian Wolf
5. All of the following Ethiopian endemic animals are on the brick of extinction, except:

A. Walia ibex
B. Black lion
C. Grevy's zebra
D. Mountain Nyala

6. Wild life conservation areas established to focus on a particular species is:
A. National park
B. Sanctuaries
C. Wild life reserve
D. Zoos

7. An area for conservation, management and propagation of Wild life are:
A. Sanctuaries
B. Wild life reserves
C. National parks
D. Zoos

8. Which of the following animal is found in Awash national Park?
A. Hamadyrs baboon
B. Gelada baboon
C. Walia ibex
D. Mountain Nyala

9. Harar wildlife sanctuary is involved in protection of:
A. Harte beest
B. Minelik's Bush buck
C. Elephant
D. Ostrich

10. Wildass, kudu, Grevy's Zebra and cheetah are all found in _____ national park.
A. Semien mountain
B. Mago
C. Rift Valley lake
D. Yangudi Rassa

Part II: Match the following endemic animals with the national park in which they are located.

Column A		Column B
1.	Walia ibex	a. Gambella
2.	Mountain Nyala	b. Bale
3.	Swayne Hartbeest	c. Semien mountains
4.	Nile perch	d. Yangudi Rasssa
5.	Wild ass	e. Maze

Sanctuaries in Ethiopia

In Ethiopia there are number of wild life sanctuaries where some species of wildlife are protected. These include:

- **Harar wildlife sanctuaries: located in Oromia region Hararge zone and it is involved in protection**
- **Kuni muktar Mountain nyala sanctuary**
- **Senkelle swaynes's hartebeest sanctuary**
- **Yabello wildlife sanctuary**
- **Didessa wildlife sanctuary**

11.5 Air

Air is a composition of different substances. It consists of different gases, microorganisms, pollutants, and others. Some of the major components of the atmospheric air are:

- Carbon dioxide
- Oxygen
- Nitrogen
- Water vapor, etc.

Among these oxygen and carbon dioxide are the most important ones.

- ***Oxygen***: is needed to oxidize food or organic substances during aerobic respiration.
- ***Carbon dioxide***: used during the process of photosynthesis as a source of carbon that is found in a glucose molecule.

Due to an increasing number of human populations, the demand of humans is increasing at an alarming rate. This has created:

- ✓ Migration of people from rural to urban areas which is contributing large in increasing of ***urbanization***.
- ✓ ***Industrialization*** which is contributing the largest proportion in the pollution on the world.

What is Pollution?

Pollution is a contamination of natural environment by harmful substances as a result of human activities. Pollution can be caused by several factors some this includes:

- ✓ Industrial wastes,
- ✓ Household wastes;
- ✓ Exhaust from vehicles;
- ✓ Release of pollutant gases in to the atmosphere, etc.

Pollutants are chemical substances that cause contamination of the air, soil and water.

Forms of air pollution

Air pollution can be caused in different forms. These include:

1. ***Global dimming:*** is a blockage or reduction of sunlight from reaching the earth's surface as a result of un burnt hydrocarbons released in to the atmosphere in the form of smoke. It is sometimes called **black carbon pollution.**
2. ***Global warming:*** is the rising of temperature on earth's surface as a result of accumulation of green house gases such as carbon dioxide and methane in to the atmosphere.

Green house gases trap heat from the sun keep the earth's surface warm. This effect is known as **green house effect.**
Examples of green house gases include:

- Mainly carbon dioxide
- Methane
- Surface ozone
- Nitrous oxide
- Chlorofluorocarbons (CFC)

The increasing level of greenhouse gases in the atmosphere minly as a result of human activities causes increase of the earth's temperature.

Some of the green house gases and their causes are discussed below.

1. Carbon dioxide

The level of carbon dioxide is increasing as a result of human activities. This involves:

- Burning of fossil fuels from cars, airplanes, and different power stations.
- Deforestation which causes destruction of forest habitat that results in the reduction of the absorption of atmospheric CO_2 during photosynthesis.

2. Methane

Methane is a green house gas that is released in to the atmosphere as a result of:

- Growth of rice in swampy conditions where methane is largely released.
- Digestive activity of cattle also produces methane. Thus, there is methane in the dung of cattle.

3. CFC (Chlorofluorocarbons)

CFC's are chemicals that cause ozone depletion. **Ozone layer depletion,** is simply the thinning (reduction) of the amount of ozone in the stratosphere.

The **ozone layer** or **ozone shield** is a region of Earth's stratosphere that absorbs most of the Sun's ultraviolet radiation.

Other pollutants

Carbon monoxide- is a pollutant that is released in to the atmosphere as a result of the burning of fossil fuels.

- It also affects human beings by reducing the oxygen carrying capacity of the blood by combining with hemoglobin (oxygen carrying substance). The poisoning of carbon monoxide can lead to death and it is difficult to detect because it has no smell or color.
- It is also among the chemicals found in a cigarette smoke.

Sulfuric oxide and nitrogen oxide

Chemicals such as sulfuric oxide and nitrogen oxide are released in the atmosphere from factories, cars and other vehicles.

- Acid rain is formed when the falling rain water reacts with oxides of sulfur and nitrogen.
- Acid rain causes several damages on the environment by:
  - Killing leaves;
  - Depletion of soil which destroys roots of trees;
  - Pollution of lakes, river, and stream water;
  - Damage on building materials by reacting with calcium carbonate.
- The natural causes of acid rain include:
  - volcanic eruption and lightening;
  - water vapor from clouds;
  - rain
  - human causes for acid rain are burning of fossil fuels from industries and vehicles;

Other effects of air pollution on humans

Air pollution can affect our health in various ways. Some of the effects are discussed below:

- chronic respiratory problems;
- lung cancer;
- heart disease;
- brain damage;
- nerve, liver and kidney problem;
- development of skin problems including skin cancer by the damage of UV- rays as a result of ozone depletion;

Prevention of Air Pollution in Ethiopia

Air pollution is a global problem. Thus all nations have a responsibility to reduce air pollution.

Just like other nations on the world, Ethiopia has a responsibility in prevention of air pollution. Here are some of the preventive measures that shall be taken as a nation in Ethiopia:

- reduction or avoiding the use of traditional fuels such as wood and animal dung;
- using non-pollutant alternative sources of energy;
- establishment of laws to control the pollution from factories;
- controlling fire on forest;
- avoiding the release of industrial chemicals in to rivers and lakes;
- educating the society on mechanisms that help to reduce air pollution.

Illustrative Examples

61. Which of the following is the effect of depletion of ozone layer?
    A. Easy access of ultraviolet radiation to humans.
    B. Intensification of global warming.
    C. High frequency of acid rain.
    D. Oxygen deficiency in the atmosphere.

    Explanation:
    Ozone large (O_3) is a blanket that prevents excessive uv – rays from reaching the earth surface.

    Answer: A

62. Which of the following is correct about the greenhouse effect?
    A. It is a natural phenomenon that maintain earth's temperature range.

B. It is the result of excess of carbondioxide in the atmosphere.
C. It is a natural phenomenon that causes heat energy to be radiated back into the atmosphere.
D. It is the result of the difference in the angle of the sun rays.

Explanation:
Green house is a glassy house which slows down the escape of heat. The excess carbon dioxide slows down the radiation of heat back into the outer space which causes **global warming** and its effect is called **greenhouse effect.**

Answer: B

63. Clearly define the following.
- National park
- Wildlife sanctuaries
- Wildlife reserve

Explanation:
- **National park** – is a large conservation area established for protection of wild animals and vegetation.
- **Wild life sanctuaries** – is a conservation area designed to provide protection for particular species of organisms.
- **Wild life reserve** – is an area designed for conservation management and propagation of wild life.

64. List out some of the wild life sanctuaries in Ethiopia with their locations and animals that are kept in them.

Explanation:

Sanctuaries	Location	Animal protected
Harar wildlife sanctuary	Oromia region Hararge zone	Elephant, Black lion
Kuni Muktar wildlife sanctuary	Oromia	Mountian Nyala
Senkelle wildlife sanctuary	Oromia and SNNPR	Hert beest
Yebello wildlife sanctuary	Borena zone Oromia region	Ostrich

65. Which of the following method is used to conserve wildlife?
A. Setting national parks and sanctuaries.

B. Shifting their habitat and niche.
C. Allowing poaching for economical use.
D. Mixing wild life with the domestic species.

Explanation: Setting wild life protection areas such as national parks, zoos and sanctuaries is very important in conservation of wild life. **Answer: A**

66. Which of the following substances increase the greenhouse effect the most?
A. Oxygen and CFS's
B. Methane and carbondioxide
C. Nitrogen and methane
D. Sulfurdioxide and nitrogen

Explanation: Methane and carbondioxide are the major greenhouse gases. **Answer: B**

67. The main pollutants involved in depleting the ozone layer are:
A. Fossil fuels
B. Industrial furnaces
C. Pesticides
D. Chlorofluoro carbons

Explanation: Chlorofluorocarbons (CFC's) are released from aerosol sprays and refrigerators. They are pollutants that deplete ozone (O_3) layer and cause ozone hole. **Answer: D**

68. Which of the following method is used to prevent air pollution?
A. Deforestation
B. Burning of fossil fuels
C. Reforestation
D. Industries

Explanation: Air pollution can be prevented by:
- Afforestation and reforestation
- Avoiding use of traditional fuels such as wood and animal dung
- Use of alternative source of energy
- Establishment of laws to control the pollution from factors
- Controlling fire on forest
- Avoiding release of industrial chemicals
- Educating the society

Answer: C

69. Which of the following gases result in global warming when accumulated in the atmosphere?
A. Hydrogen
B. Nitrogen
C. Carbon dioxide
D. Oxygen

Explanation:

Global warming is an increase (rise) of earth's temperature which is caused due to the accumulation of greenhouse gases like **carbon dioxide** and **methane**.

70. Pollution by chlorofluorocarbons is caused: **Answer: C**

A. acid rain
B. Volcanic eruption
C. global warming
D. Ozone hole

Explanation: Refer example number 67.

71. Which of the following is Not cause for air pollution? **Answer: D**

A. Cars and vehicles
B. Afforestation
C. Industries
D. Deforestation

Explanation: Refer example number 68.

72. The greenhouse effect is: **Answer: B**

A. The result of the difference in the angle of the sun's rays.
B. A natural phenomenon that causes heat energy to radiate back to the atmosphere.
C. The result of excess carbondioxide in the atmosphere.
D. A natural phenomenon that maintains earth's temperature range.

Explanation:
Greenhouse effect is caused by the effect of gases like carbondioxide and methane to trap heat from the sun and keeps the earth's surface warm.

Answer: C

73. Deforestation clears plants from the environment that trigger global warming. How does deforestation facilitate global warming?

A. Deforestation facilitates the accumulation of CO_2 in the air that leads to global warming.
B. Deforestation is important to reduce CO_2 from the environmental air.
C. Deforestation facilitates accumulation of excess O_3 in the air leads to global warming.
D. Deforestation facilitates accumulation of excess O_2 in the air that leads to global warming.

Explanation: Deforestation is the act of removing or cutting the vegetation cover in an area. Thus removal of vegetation facilitates

the accumulation of CO_2 in the atmosphere which can be used by plants for photosynthesis. **Answer: A**

74. Which of the following activities cause global warming?
   A. Reducing the number of livestock.
   B. Reducing the amount of rice farms.
   C. Burning of fossil fuels.
   D. Afforestation of the habitat.

   Explanation: Global warming can be caused by:
   - Release of toxic chemicals (pollutants) into the atmosphere
   - During of fossil fuels
   - Destruction (deforestation) of forest habitat **Answer: C**

75. Explain the meanings of:
   - Global warming
   - Global warming
   - Acid rain

   Explanation:
   - **Global warming**: is the reduction of sun light from reaching the earth's surface as a result of unburnt hydrocarbons in the atmosphere.
   - **Global warming**: is the rise of earth's temperature, as a result of accumulation of greenhouse gases in the atmosphere.
   - **Acid rain**: is a rain that forms when the rain water reacts with oxides of sulphur and nitrogen.

Posttest – 4

Part I: Choose the best answer from the given alternatives.

1. Which of the following is the effect of depletion of ozone layer:
   A. Oxygen deficiency in the atmosphere.
   B. Easy access of ultra violet radiation to humans.
   C. Intensification of global warming.
   D. High frequency of acid rain.
2. Burning of fossil fuels causes air pollution by:
   A. Producing and releasing nitrogen dioxide into the air.
   B. Releasing burnt hydrocarbon particles into the air.
   C. Releasing unburnt hydro carbon particles into the air.
   D. Producing and releasing carbon dioxide into the air.
3. Acid rain is formed as a result of one of the following pollutant. Which one is it?

A. Carbondioxide
B. Chlorofluoro carbon
C. Sulfur dioxide
D. Methane

4. Global warming is caused by:
A. CFC and nitrous oxide
B. Methane and Sulfur dioxide
C. Carbondioxide and methane
D. Carbon monoxide and oxygen

5. A pollutant gas that reduces oxygen carrying capacity of the blood is:
A. Carbon di oxide
B. Sulfur oxide
C. Carbon monoxide
D. Methane

6. Ozone hole is formed as a result of the release of:
A. Chlorofluoro carbons
B. Green house gases
C. Carbon dioxide
D. Methane

7. Which of the following human activity plays role in increasing global warming?
A. Afforestation
B. Reforestation
C. Cultivation of tree
D. Deforestation

8. Excessive UV-rays are protected from reaching the earth's surface by:
A. Ozone
B. Greenhouse gases
C. CFC's
D. Acid rain

9. Digestive activity of cattles results in the release of one of the following pollutants:
A. Carbon monoxide
B. Methane
C. Carbondioxide
D. CFS's

10. Blockage of reduction of sunlight from reaching the earths surface is called:
A. Global dimming
B. Global warming
C. Ozone hole
D. Greenhouse effect

Supplementary Exercise on Unit Eleven

1. One of the following natural resource cannot be easily replaced, once it has been sued:
A. Water B. Tree C. gold D. wind

2. Which of the following cannot produce a renewable form of energy?

A. Carbondioxide
B. Chlorofluoro carbon
C. Sulfur dioxide
D. Methane

4. Global warming is caused by:
A. CFC and nitrous oxide
B. Methane and Sulfur dioxide
C. Carbondioxide and methane
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Supplementary Exercise on Unit Eleven

1. One of the following natural resource cannot be easily replaced, once it has been sued:
A. Water B. Tree C. gold D. wind

2. Which of the following cannot produce a renewable form of energy?

A. Petroleum B. Sunlight C. wind D. water

3. The lose of species from a particular habitat is:
   A. Overgrazing
   B. Extinction
   C. Deforestation
   D. Afforestation
4. Biodiversity is measured
   A. The area covered by vegetation
   B. The topography of the forest habitat
   C. The variety of species it contains.
   D. All are possible
5. All of the following are importance of biodiversity, except
   A. The balance in an ecosystem
   B. To develop genetic fiversity of crops and animals
   C. To have areas with admirable and wonderful biodiversity
   D. To have monocultures with a vegetation of single species for industries.
6. Which of the following is importance of monoculture:
   A. Its attractiveness for recreational purpose
   B. It decreases risk of diseases to spread.
   C. It is important for cultivation of industrial raw materials.
   D. It supports wide range of plants and animals.
7. Biodiversity of a given habitat is not lost by:
   A. Deforestation
   B. Environmental pollution
   C. Overgrazing
   D. Afforestation
8. All of the following can be regarded as a way to conserve the biodiversity **except**:
   A. Establishment of legal actions
   B. Reduction of herbivores by predators
   C. Release of pollutants
   D. Establishing wild life conservation areas
9. When we say Ethiopia is a county with diversity of vegetation, it means to say:
   A. Large area of Ethiopian land is covered by vegetation
   B. Ethiopian vegetation cover is at satisfying level.

C. Great emphasis was given to protect Ethiopian vegetation.
D. Ethiopia has variety of species of vegetation.

10. The importance of vegetation (plant) to an area is:
A. To absorb atmospheric carbpndioxide
B. Serves as a habitat for various types of animals
C. Releases oxygen in to the atmosphere
D. All are possible answers.

11. Which of the following is not a cause of deforestation?
A. Overgrazing by domestic animals.
B. Release of pollutants from factories and vehicles
C. Production of timber
D. Planting of new trees to replace harvested one.

12. Which of the following is **NOT** an endemic plant species of Ethiopia?
A. Eragrastis teff
B. Enset
C. ficus vista
D. eucalyptus

13. Among the list of Emperors given below, who is highly accredited for conservation of vegetation?
A. Zera Yakob
B. Minellik II
C. Haileseliasie
D. Taytu Betul

14. Which of the following mechanisms should not be used to conserve Ethiopian vegetation cover
A. Educating people
B. Afforestation
C. Development of monocultures
D. Establishing national parks

15. Wild animals serve to maintain the natural balance of an ecosystem. Which of the following examples best explains the above statement?
A. Overgrazing of a land by harbivores
B. Controlled number of herbivores by predators
C. Reduction of number of herbivopres as it reaches carry ingcapacity
D. Deforestation of vegetations by wild animals.

16. In Ethiopia due to expansion of sugar factories, large forest vegetation areas are replaced by a sugar cane plants. This is example best described:
   A. Deforestation
   B. Monoculture
   C. conservation
   D. Biodiversity

17. Which of the following is not considered to be the importance of wildlife resource?
   A. Commercial value
   B. Educational value
   C. Maintenance of natural balance in an ecosystem
   D. Reducing the level of carbondioxide in the atmosphere

18. Which of the following is Ethiopian Endemic mammalian species?
   A. Thick-billed raven
   B. Wattled ibis
   C. Mountain Nyala
   D. White – collared pigeon

19. Which of the following is NOT a human effect on wild life habitat?
   A. Destruction of forest habitat
   B. Uncontrolled hunting
   C. Human setting
   D. Volcanic eruption

20. Among the following wild species of Ethiopia, which ones are on the brick of extinction?
   A. Grevy's zebras
   B. Gelada baboon
   C. swayne's hatte beest
   D. Menelik's Bush buck

21. A conservation area designed to give a maximum protection to wild animals is:
   A. Wildlife sanctuaries
   B. Zoos
   C. wild life reserve
   D. National parks

22. A highly protected area that involves control of human settlement, hunting and capturing wild animals is:
   A. Wildlife reserve
   B. National parks
   C. zoo
   D. wildlife sanctuaries

23. A national park that consists of wild animals like hippos, roan antelope and crocodiles is:
   A. Bale mountains
   B. Gambella
   C. Omo
   D. Maze

24. A national park that consists of animals like Swayne hart beast is:
   A. Maze
   B. Rift valley lakes
   C. Bale Mountains
   D. Awash

25. _____ Sanctuary is dedicated to protection of endemic mountain Nyala.
   A. Harar wildlife sanctuary
   B. Kuni muktar sanctuary
   C. Senkelle sanctuary
   D. Yabello wild life sanctuary.

26. Which of the following gases is responsible for global warming?
   A. Gulphur oxide
   B. Nitrogen oxide
   C. carbondioxide
   D. carbon monoxide

27. A gas that reduces the oxygen carrying capacity of the blood is:
   A. Nitrogen oxide
   B. Cabonmonoxide
   C. carbondioxide
   D. methane

28. Which of the following gases have green house effect?
   A. Carbon monoxide and nitrogen oxide
   B. CFC'S and carbon monoxide
   C. Carbondioxide and sulphur oxide
   D. Methane and carbondioxide

29. Which of the following is not caused as an effect acid rain?
   A. Killing of plant leaves
   B. Soil depletion
   C. global warming
   D. damage on building

30. The pollutant that increases the risk of skin cancer by its effect on ozone layer is:
   A. Sulphur oxides
   B. CFC'S
   C. carbondioxide
   D. Methane

31. Baro river is the part of
A. Gambela national park C. Awash national park
B. Mago national park D. Yangudirasa

Answers to Post Test

Posttest – I

Part I

1.	B	2.	C	3.	B	4.	C	5.	B
6.	C	7.	A	8.	B	9.	A	10.	D
11.	A	12.	A	13.	B				

Part II

14.

Renewable natural resources	Non-renewable natural resources
- Water - Animal - Air - Vegetation	- Coal - Petroleum - diamond

15.

- **Extinction** – is elimination of species from the land due to natural catastrophes or manmade factors.
- **Monoculture** – replacement of the biodiversity only by a single – vegetation species.
- **Biodiversity** – is a wealth of species in a given habitat.

Post Test – 3

Part I

1.	C	2.	A	3.	A	4.	C	5.	B
6.	B	7.	B	8.	B	9.	C	10.	D

Part II

1.	C	2.	B	3.	E	4.	A	5.	D

Post Test – 4

Part – I

1.	B	2.	B	3.	C	4.	C	5.	C
6.	A	7.	D	8.	A	9.	B	10.	A

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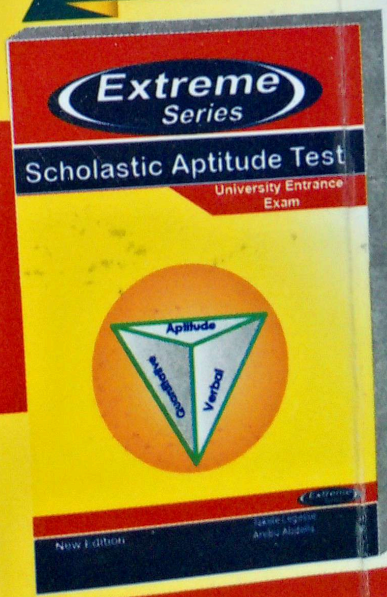

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